9/11 ch.1.2

$f\circ g(x) = f(g(x))$: composition of $f(\cdot)$ and $g(\cdot)$

positive measure：正向角 (逆時針)

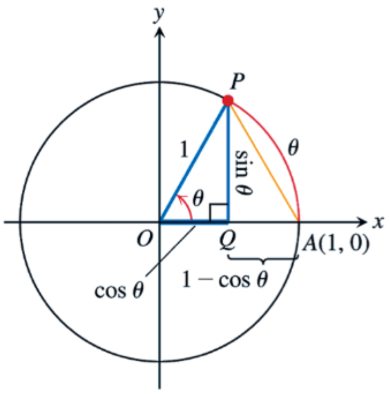
negative measure：負向角 (順時針)

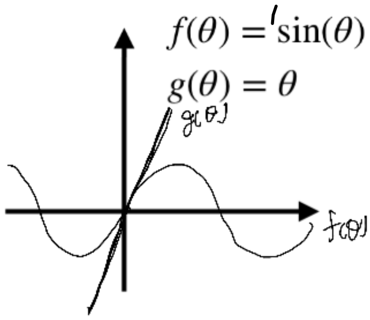
adjacent $\quad$ $\sin = \frac{y}{h}$ $\quad$ csc $\cdot$ $\frac{}{h}$

opposite $\quad$ $\cos = \frac{x}{h}$ $\quad$ sec

hypotenuse $\quad$ $\tan = \frac{y}{x}$ $\quad$ cot

Characteristic of $\sin(\theta)$





$f(\theta) = \sin(\theta)$

$g(\theta) = \theta$

- Consider a unit circle (with radius 1)
- The arc length of the central angle $0 \le \theta \le \pi/2$ is previously obtained as

$$\widehat{PA} = \theta$$

$\sin^2\theta + 1 - 2\cos\theta + \cos^2\theta$

$2 - 2\sin\theta\cos\theta$

- In addition,

$$PA \ge \overline{PA} = \sqrt{\sin^2\theta + (1-\cos\theta)^2}$$
$$\ge \sqrt{\sin^2(\theta)} = \sin\theta$$

- We thus conclude that

$$\sin(\theta) \le \theta \ \text{ for } \ 0 \le \theta \le \pi/2$$

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Exponential functions：指數函數 : $f(x) = c\cdot a^x$

Iverse functions: A function can be reasoned its input by its output.

One-to-One functions : Every output results from uniquely one input, then the answer is positive.
For any two distinct input x_1 and x_2 We have $h(x_1) \neq h(x_2)$

Inverse

> **Definition.** If $h(\cdot)$ is one-to-one, then the inverse of $h(\cdot)$ exists and is defined by $h^{-1}(\cdot)$

Inverse graph

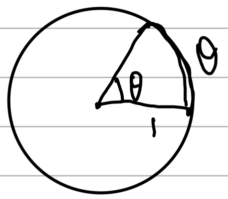
> The graphs of $g(x)$ and $g^{-1}(x)$ are reflections of each other about the line $y = x$!

Logarithm function

Domain restriction

Inverse of Trigonometric functions

The inverse of (the properly resticted) $\sin x$ is defined as $\sin^{-1}x = \arcsin x$



$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ arcsin. The arc whose $\sin x$ is $\frac{\sqrt{3}}{2}$

$\theta = \frac{\pi}{3}$

Relation between the inverse of tringonometric functions.

- $\cos^{-1}x + \sin^{-1}x = \frac{\pi}{2}$ $\quad 1 \geq x \geq 0$

Ch.2.2

Indentity function: Direcly output the result from input.

Unit step function: $\begin{cases} 1, & x<0 \\ \frac{1}{2}, & x=0 \\ 1, & x>0 \end{cases}$